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Analysis of Electrochemical Impedance Spectroscopy Data Using the Distribution of Relaxation Times: A Bayesian and Hierarchical Bayesian Approach

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ABSTRACT

Electrochemical impedance spectroscopy (EIS) is one of the most important experimental techniques employed in electrochemistry because it can be used to deconvolve physico-chemical phenomena occurring at disparate timescales. Unfortunately, the analysis of EIS data is frequently challenging since it can require the selection of *ad hoc* equivalent circuits. The distribution of relaxation times (DRT) method is complementary to the approach of fitting equivalent circuits because the DRT maps the EIS data onto a function containing the timescale characteristics of the system under study. While conceptually simple, the DRT cannot be obtained by simple minimization of the least squares because the corresponding optimization problem is ill posed. Regularization methods, such as ridge/Tikhonov or Lasso regression, add a penalty term to the least squares minimization problem enabling the DRT deconvolution. In this work, we show that such regularization methods may be understood in a Bayesian context. For example, ridge/Tikhonov regression implicitly encapsulates the prior insight that the derivatives of the DRT are regular. We use this Bayesian approach as a starting point to extend the DRT regularization by considering frequency dependent oscillation levels. This approach is shown to be more robust with respect to both discontinuities and over smoothing than typical regularized DRT methods. Furthermore, the Bayesian approach is versatile and may be extended to include more informative priors.

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1. Introduction

Electrochemical impedance spectroscopy (EIS) is one of the key techniques used in electrochemistry [1–5] and has been utilized in many areas including fuel cells [6–10], batteries [11–14], sensors [15–17], capacitors [18], dielectrics [19,20], electrochemical coating [21,22], imaging [23,24], and biology [25,26], just to list a few applications. The EIS is particularly useful in these fields because it is conducted over a broad range of frequencies allowing the deconvolution of physic-chemical phenomena characterized by disparate timescales [27].

The EIS data are acquired by applying small voltage (or current) perturbation to an electrochemical system so as to measure the corresponding current (or voltage) [2]. This is repeated at various frequencies to obtain the EIS spectrum. The latter is typically understood as the ratio between the voltage and the current in frequency space [5]. Namely, it is a complex-valued function defined as the Fourier transform of the potential $v(t)$ divided by the Fourier transform of the current $i(t)$

$$Z(f) = \frac{V(f)}{I(f)} \tag{1}$$

where $V(f) = \mathcal{F}[v(t)](f)$ and $I(f) = \mathcal{F}[i(t)](f)$ and where we take the unitary Fourier transform definition: $\mathcal{F}[h(t)](f) = \int_{-\infty}^{\infty} h(t)\exp(-2\pi i f t)\,dt$. The experimental impedance is then used to understand the physico-chemical properties of the system under study. For this purpose having a reliable model is critical because it aids the experimental data interpretation. Typical EIS models consist of a collection of elementary circuits, e.g., resistors, capacitors, constant phase elements, and Warburg circuits placed in series or in parallel. In spite of notable exceptions [27–38], the circuits are often selected *ad hoc* so as to follow physical intuition and the principle of parsimony [27,39,40]. Furthermore, such equivalent circuits may not be unique, in that several of them may fit the data equally well. This comes to the detriment of the physical insight that one can obtain from the EIS experiments.

One way to bypass the lack of uniqueness of problem-specific equivalent circuits (and therefore complement the analysis of the EIS) is to use the distribution of relaxation times (DRT) method

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[10,14,41–54], which models the impedance as $Z_{\mathrm{DRT}}(\boldsymbol{x}, f)$, where $\boldsymbol{x}$ is an unknown and possibly large vector. In turn $\boldsymbol{x}$ can be mapped onto the relaxation characteristics that characterize the electrochemical system under study. The entries of $\boldsymbol{x}$ are obtained by minimizing the sum of the absolute value of the residuals computed between the experimental data $Z_{\exp}(f)$ and $Z_{\mathrm{DRT}}(\boldsymbol{x}, f)$ at the experimental frequencies f_n

$$S(\boldsymbol{x}) = \sum_{n=1}^{N} \left[w'_n \left(Z'_{\exp}(f_n) - Z'_{\mathrm{DRT}}(\boldsymbol{x}, f_n) \right)^2 + w''_n \left(Z''_{\exp}(f_n) - Z''_{\mathrm{DRT}}(\boldsymbol{x}, f_n) \right)^2 \right] \tag{2}$$

where Z' and Z'' indicate the real and imaginary part of the impedance respectively and w'_n and w''_n are suitable weights. If $\boldsymbol{x}$ has size close to N, the minimization of (2) is ill posed yielding solutions highly dependent on the experimental error. There are a number of ways to circumvent this problem. Researchers have employed Fourier transformation and filtering [41,47], Monte Carlo techniques [43,44], maximum entropy methods [55–57], and advanced evolutionary programming [50–52,58]. One particularly popular method consists in minimizing the following expression [10,14,59–66]:

$$S(\boldsymbol{x}) = \sum_{n=1}^{N} \left[w'_n \left(Z'_{\exp}(f_n) - Z'_{\mathrm{DRT}}(\boldsymbol{x}, f_n) \right)^2 + w''_n \left(Z''_{\exp}(f_n) - Z''_{\mathrm{DRT}}(\boldsymbol{x}, f_n) \right)^2 \right] + \lambda P(\boldsymbol{x}) \tag{3}$$

where the second term on the left hand side is the product of a function $P(\boldsymbol{x})$, a penalty, and a positive parameter λ. The penalty can be, as in ridge (or Tikhonov) regression, the norm of the second derivatives of the DRT obtained from $\boldsymbol{x}$.

In this article we aim at answering two open questions regarding DRT analysis:

1. Can the penalty term in the minimization problem (3) be understood using statistics?
2. Can we find a statistically motivated method to extend ridge DRT so that the level of regularization (as expressed by λ in (3)) can vary across the timescales?

We show that the regularized DRT can be derived from Bayesian statistics arguments [67]. In other words, the term $\lambda P(\boldsymbol{x})$ in (3) encapsulates the prior physical information available on the DRT. For example, ridge regression provides the (prior) information that the qth-order derivative of the DRT is distributed as a Gaussian random variable with standard deviation $\frac{1}{\sqrt{\lambda}}$. Therefore, the smaller λ is, the larger the oscillations are expected to be. Conversely, a large λ implies that one expects much smaller oscillations. This answers the first question.

The simple Bayesian approach outlined above assumes, however, that the level of regularization is uniform throughout the entire frequency spectrum. Equivalently, the prior implies that the same oscillation rates in the DRT are expected to occur across all timescales. In order to select a local λ and yet take only a limited number of tunable parameters, a hierarchy of Bayesian priors need to be used [68]. In this hierarchical approach (we have a prior of the prior itself), the DRT and the optimal penalty level are found simultaneously. This addresses the second question.

More broadly, the Bayesian framework proposed in this article serves as a starting point for extending DRT regression and for improving the interpretation of the DRT spectra.

2. Theory

2.1. The DRT Method

As outlined in the introduction, the DRT method assumes that the response of the electrochemical system under study is obtained from a distribution of relaxations (see Appendix A for details). Thus, the impedance can be written as

$$Z_{\mathrm{DRT}}(f) = R_\infty + \int_0^\infty \frac{g(\tau)}{1 + i2\pi f\tau} \mathrm{d}\tau \tag{4}$$

where R_∞ and $g(\tau)$ are both non negative and where the DRT subscript is used to emphasize that $\mathscr{F}^{-1}[Z_{\mathrm{DRT}}(f)](t)$ is a sum of decaying exponentials. Since many electrochemical experiments are conducted with a given number of points per decade, the (4) can be more conveniently rewritten as

$$Z_{\mathrm{DRT}}(f) = R_\infty + \int_{-\infty}^\infty \frac{\gamma(\ln\tau)}{1 + i2\pi f\tau} d\ln\tau \tag{5}$$

where $\gamma(\ln\tau) = \tau g(\tau) \geq 0$. We will use (5) in the remainder of the article.

The main goal of the DRT analysis is to obtain an estimate of $\gamma(\ln\tau)$. In order to do that, we first need to approximate $\gamma(\ln\tau)$ and $Z_{\mathrm{DRT}}(f)$ using a suitable discretization. Subsequently, we estimate the discrete approximation using regression. The discretization of $\gamma(\ln\tau)$ can be obtained by expanding the DRT over a given finite basis $\mathscr{B} = \left\{\psi_1(\ln\tau), \psi_2(\ln\tau), \ldots, \psi_M(\ln\tau)\right\}$ as [69]

$$\gamma(\ln\tau) = \sum_{m=1}^{M} x_m \psi_m(\ln\tau) + e^{\mathrm{discr}}(\ln\tau) \tag{6}$$

where the x_m's are scalars and where $e^{\mathrm{discr}}(\ln\tau)$ is the discretization error. The latter depends on the basis $\mathscr{B}$ chosen and on the particular function $\gamma(\ln\tau)$. By plugging (6) into (5), we can write the following vector equation

$$\boldsymbol{Z}_{\mathrm{DRT}} = R_\infty \mathbf{1} + \boldsymbol{A}'\boldsymbol{x} + i\boldsymbol{A}''\boldsymbol{x} + \boldsymbol{e}^{\mathrm{approx}} \tag{7}$$

where $(\boldsymbol{Z}_{\mathrm{DRT}})_n = Z_{\mathrm{DRT}}(f_n)$ with $1 \leq n \leq N$, $\mathbf{1}$ is a vector with N entries all equal to 1, $\boldsymbol{x} = (x_1, x_2, \ldots, x_M)^T$, $\boldsymbol{A}'$ and $\boldsymbol{A}''$ are real $N \times M$ matrices, and $(\boldsymbol{e}^{\mathrm{approx}})_n$ is the error made in approximating (at the frequency f_n) the DRT (5) using the first 3 terms on the right hand side of equation (7). We emphasize that M, the dimension of the basis, and N, the total number of experimental points, need not be identical. Further, we will consider that our model is simply $\boldsymbol{Z}_{\mathrm{DRT}} = R_\infty \mathbf{1} + \boldsymbol{A}'\boldsymbol{x} + i\boldsymbol{A}''\boldsymbol{x}$. The $\boldsymbol{x}$ of the expansion (6) can then be obtained via regularized regression by solving the following problem with respect to $\boldsymbol{x}$ [53]

$$\boldsymbol{x} = \underset{\boldsymbol{x} \geq 0}{\operatorname{argmin}} \left[\left\| \boldsymbol{\Omega}' \left(R_\infty \mathbf{1} + \boldsymbol{A}'\boldsymbol{x} - \boldsymbol{Z}'_{\exp} \right) \right\|^2 + \left\| \boldsymbol{\Omega}'' \left(\boldsymbol{A}''\boldsymbol{x} - \boldsymbol{Z}''_{\exp} \right) \right\|^2 + \lambda P(\boldsymbol{x}) \right] \tag{8}$$

The first two terms are the sum of the residuals (the measure of the distance between the data and the model) weighted in accordance to the matrices $\boldsymbol{\Omega}'$ and $\boldsymbol{\Omega}''$[i]. If we set $P(\boldsymbol{x}) = \left\| \boldsymbol{L}\boldsymbol{x} \right\|^2$, where $\boldsymbol{L}$ is a suitable qth order differentiation matrix, we obtain the ridge DRT.

[i] In reference to (2), $\boldsymbol{\Omega}' = \mathrm{diag}\left(\frac{1}{\sqrt{w'_1}}, \frac{1}{\sqrt{w'_2}}, \ldots, \frac{1}{\sqrt{w'_N}} \right)$ $\boldsymbol{\Omega}'' = \mathrm{diag}\left(\frac{1}{\sqrt{w''_1}}, \frac{1}{\sqrt{w''_2}}, \ldots, \frac{1}{\sqrt{w''_N}} \right)$.

If instead $P(\boldsymbol{x}) = \|\boldsymbol{x}\|_1 = \sum_{m=1}^{M} |x_m|$, we recover the Lasso DRT [53]. As it will be discussed in the following Section, the penalty term can be understood as a Bayesian prior on the DRT.

2.2. A Bayesian Perspective on DRT

We will suppose that a given EIS experiment is a realization of the following stochastic process:

$$\boldsymbol{Z}_{\exp} = R_\infty 1 + \boldsymbol{A}'\boldsymbol{x} + i\boldsymbol{A}''\boldsymbol{x} + \boldsymbol{E}' + i\boldsymbol{E}'' \tag{9}$$

where $\boldsymbol{E}'$ and $\boldsymbol{E}''$ are independent random variables. The centerpiece of our analysis is Bayes formula, which, in the context of DRT, can be written as [67,70]

$$p\left(\boldsymbol{x}|\boldsymbol{Z}'_{\exp}, \boldsymbol{Z}''_{\exp}\right) p\left(\boldsymbol{Z}'_{\exp}, \boldsymbol{Z}''_{\exp}\right) = p(\boldsymbol{x})\, p\left(\boldsymbol{Z}'_{\exp}, \boldsymbol{Z}''_{\exp}|\, \boldsymbol{x}\right) \tag{10}$$

where $p(\cdot)$ is a probability distribution function (pdf) of the corresponding random variable in brackets and where the symbol $|$ indicates "conditioned to" or "given". For example, $p\left(\boldsymbol{Z}'_{\exp}, \boldsymbol{Z}''_{\exp}|\, \boldsymbol{x}\right)$ (the likelihood) is the pdf of experimental outcome given $\boldsymbol{x}$ or conditioned to a certain $\boldsymbol{x}$ or equivalently assigned an approximation of $\gamma(\ln\tau)$. The logic of this Bayesian analysis is illustrated in Fig. 1. In contrast to the frequentist approach, $\boldsymbol{x}$ is interpreted as a random variable rather than a fixed true value that needs to be estimated. To that end, we will aim at constructing the pdf of $\boldsymbol{x}$ given the $\boldsymbol{Z}_{\exp} = \boldsymbol{Z}'_{\exp} + i\boldsymbol{Z}''_{\exp}$ (or equivalently given $\boldsymbol{Z}'_{\exp}, \boldsymbol{Z}''_{\exp}$) and the prior information available to the experimenter as provided by the pdf $p(\boldsymbol{x})$. We will assume that $\boldsymbol{E}'$ and $\boldsymbol{E}''$ are normally distributed, namely $\boldsymbol{E}' \sim \mathcal{N}\left(0, \boldsymbol{\Sigma}'\right)$ and $\boldsymbol{E}'' \sim \mathcal{N}\left(0, \boldsymbol{\Sigma}''\right)$. If we further assume that the all entries of $\boldsymbol{E}'$ and $\boldsymbol{E}''$ are independent, then $\boldsymbol{\Sigma}'$ and $\boldsymbol{\Sigma}''$ are diagonal matrices, i.e., $\boldsymbol{\Sigma}' = \mathrm{diag}\left(\sigma'^2_1, \sigma'^2_2, \ldots, \sigma'^2_N\right)$ and $\boldsymbol{\Sigma}'' = \mathrm{diag}\left(\sigma''^2_1, \sigma''^2_2, \ldots, \sigma''^2_N\right)$. Furthermore, we can easily find the likelihood by noting that the residuals on the real and on the imaginary part of the DRT model impedance, r'_j and r''_j respectively, are normally distributed; specifically,

$$r'_n = R_\infty + \left(\boldsymbol{A}'\boldsymbol{x} - \boldsymbol{Z}'_{\exp}\right)_n = E'_n \sim \mathcal{N}\left(0, \sigma'^2_n\right) \tag{11a}$$

$$r''_n = \left(\boldsymbol{A}''\boldsymbol{x} - \boldsymbol{Z}''_{\exp}\right)_n = E''_N \sim \mathcal{N}\left(0, \sigma''^2_n\right) \tag{11b}$$

The last two equations give the likelihood, the pdf of the experimental outcome given the model parameters (or equivalently given $\boldsymbol{x}$), which can be written as

$$p\left(\boldsymbol{Z}'_{\exp}, \boldsymbol{Z}''_{\exp}|\, \boldsymbol{x}\right) = p\left(\boldsymbol{Z}'_{\exp}|\, \boldsymbol{x}\right) p\left(\boldsymbol{Z}''_{\exp}|\, \boldsymbol{x}\right) \propto \exp\left(-\frac{1}{2}\sum_{n=1}^{N}\frac{r'^2_n}{\sigma'^2_n} - \frac{1}{2}\sum_{n=1}^{N}\frac{r''^2_n}{\sigma''^2_n}\right) \tag{12}$$

We note that (12) is the typical starting point of maximum likelihood estimation and nonlinear least squares regression as applied in the analysis of EIS equivalent circuits, see Fig. 1. This is because by maximizing (12) one recovers the least square problem (2) since $\boldsymbol{\Omega}' = \mathrm{diag}\left(\frac{1}{\sigma'_1}, \frac{1}{\sigma'_2}, \ldots, \frac{1}{\sigma'_N}\right)$ and $\boldsymbol{\Omega}'' = \mathrm{diag}\left(\frac{1}{\sigma''_1}, \frac{1}{\sigma''_2}, \ldots, \frac{1}{\sigma''_N}\right)$. The prior $p(\boldsymbol{x})$ requires some subjective input, which can be based on our physical insight. Therefore, we can postulate that all entries of $\boldsymbol{x}$ are non-negative, see Appendix A, and that adjacent values of $\boldsymbol{x}$ do not vary too much. For example, if we use a piecewise linear approximation these requirements can be enforced by assuming that $x_j \approx x_{j-1}$. This may be written as $x_j - x_{j-1} = \nu_j$ for $1 < j \le N$ where ν_j is some random error. By noting that $\left.\frac{d\gamma}{d\ln\tau}\right|_{\frac{x_j+x_{j-1}}{2}} \approx \frac{x_j - x_{j-1}}{\Delta\ln\tau} = \frac{1}{\Delta\ln\tau}\nu_j$, we can also write that the first order derivative approximation is distributed as a selected random variable. For example, we can assume that $\frac{1}{\Delta\ln\tau}\nu_j$ is a Gaussian random variable with mean zero and some fixed standard deviation.

In general, we can take that the approximation $\boldsymbol{Lx}$ of the qth order derivative of $\gamma(\ln\tau)$ is a random variable with prescribed pdf, e.g. $(\boldsymbol{Lx})_j \sim \mathcal{N}\left(0, \frac{1}{\lambda}\right)$. This assumption implies that we do not foresee

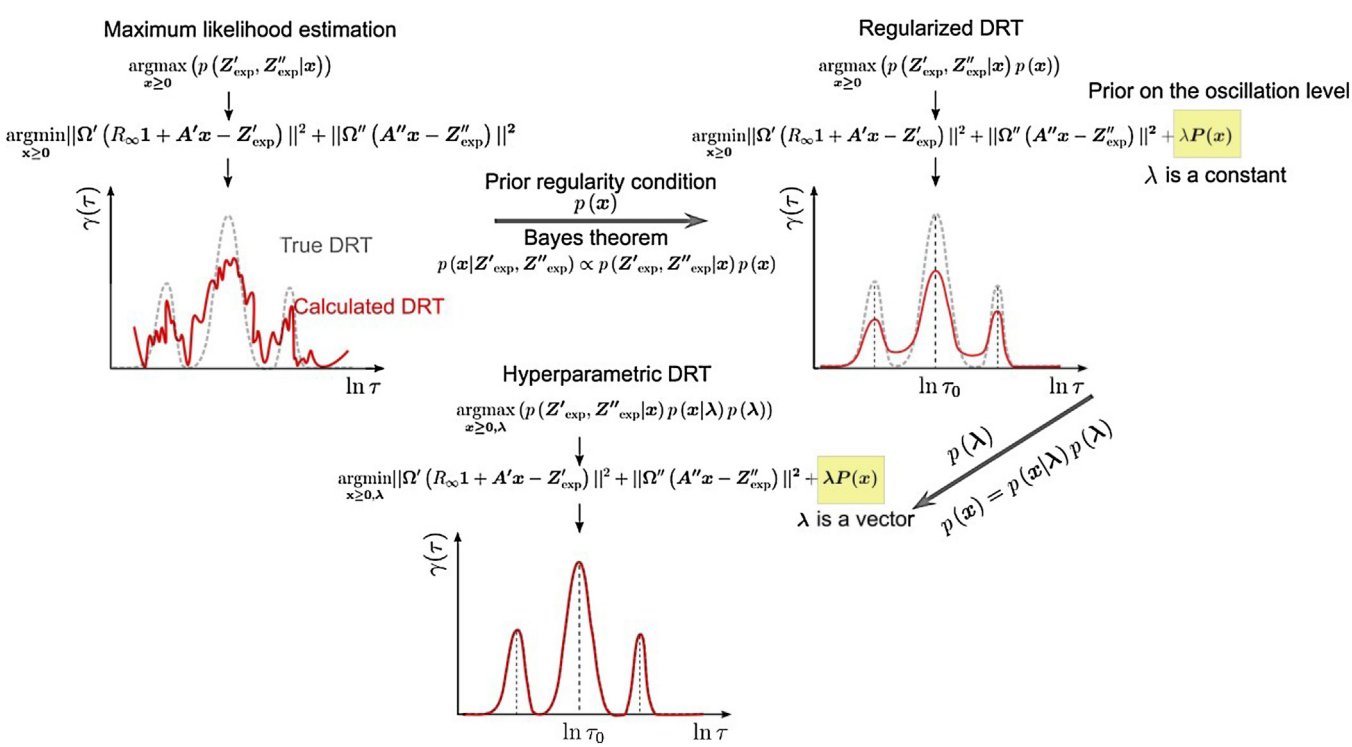

Fig. 1. Schematic depiction of the ridge and hierarchical DRT regression in relation to Bayesian statistics.

that the absolute value of the qth order derivative will be greater than $\frac{3}{\sqrt{\lambda}}$ (this will occur with <1% probability). If we further assume that the DRT response cannot be negative (this comes from physical arguments as shown in Appendix A), then we can assign the following prior to $\boldsymbol{x}$

$$p(\boldsymbol{x}) \propto \mathbb{1}(\boldsymbol{x} \geq 0) \exp\left(-\frac{\lambda}{2}\left\|\boldsymbol{L}\boldsymbol{x}\right\|^2\right) \tag{13}$$

where $\mathbb{1}(\boldsymbol{x} \geq 0)$ is the indicator function (1 if all $x_j \geq 0$ and 0 otherwise) [70]. This insight gives the posterior density, i.e., the expected pdf of $\boldsymbol{x}$ given our (prior) physical insight and the experimental data $\boldsymbol{Z}_{\exp} = \boldsymbol{Z}'_{\exp} + i\boldsymbol{Z}''_{\exp}$. In short we obtain that, see Fig. 1,

$$p\left(\boldsymbol{x}|\boldsymbol{Z}'_{\exp}, \boldsymbol{Z}''_{\exp}\right) \propto \mathbb{1}(\boldsymbol{x} \geq 0) \exp\left(-\frac{1}{2}\sum_{n=1}^{N}\frac{r'^2_n}{\sigma'^2_n} - \frac{1}{2}\sum_{n=1}^{N}\frac{r''^2_n}{\sigma''^2_n} - \frac{\lambda}{2}\left\|\boldsymbol{L}\boldsymbol{x}\right\|^2\right) \tag{14}$$

We can conveniently rewrite (14) as

$$p\left(\boldsymbol{x}|\boldsymbol{Z}'_{\exp}, \boldsymbol{Z}''_{\exp}\right) \propto \mathbb{1}(\boldsymbol{x} \geq 0) \exp\left(-\frac{1}{2}\updownarrow_{\text{ridge}}(\boldsymbol{x}, \lambda)\right) \tag{15}$$

where $\updownarrow_{\text{ridge}}(\boldsymbol{x}, \lambda)$ is the posterior negative log-likelihood (we use the subscript "ridge" to emphasize that this is ridge regression) given by

$$\updownarrow_{\text{ridge}}(\boldsymbol{x}, \lambda) = \sum_{n=1}^{N}\frac{r'^2_n}{\sigma'^2_n} + \sum_{n=1}^{N}\frac{r''^2_n}{\sigma''^2_n} + \lambda\left\|\boldsymbol{L}\boldsymbol{x}\right\|^2 \tag{16}$$

If we maximize the posterior likelihood $p\left(\boldsymbol{x}|\boldsymbol{Z}'_{\exp}, \boldsymbol{Z}''_{\exp}\right)$ with the constraint that $\boldsymbol{x} \geq 0$, then we obtain $\boldsymbol{x}_{\text{MAP}}$, the maximum *a posteriori* (MAP) estimate of $\boldsymbol{x}$ [71]. Naïvely, the $\boldsymbol{x}_{\text{MAP}}$ can be interpreted as the "most" likely $\boldsymbol{x}$, given the data $\boldsymbol{Z}_{\exp}$, and the prior physical knowledge expressed in $p(\boldsymbol{x})$. We note that the maximization of the posterior (15) is equivalent to the minimization of the negative log-likelihood (16) multiplied by two. In turn, this minimization corresponds to the typical ridge regression problem (8)

$$\boldsymbol{x}^{\text{MAP}}_{\text{ridge}} = \underset{\boldsymbol{x}\geq 0}{\text{argmin}}\updownarrow_{\text{ridge}}(\boldsymbol{x}, \lambda) \tag{17}$$

We note that λ is fixed as a consequence of having chosen the prior (13). We also note that the larger λ is the more centered towards 0 the norm of $\boldsymbol{Lx}$ will be. In other words, since we expect small qth derivatives, this prior information (or "belief") will reflect into a flattened estimated $\boldsymbol{x}^{\text{MAP}}_{\text{ridge}}$. This prior will also likely imply a large bias with respect to the ideal $\boldsymbol{x}$. Conversely, a small λ implicitly allows the derivatives $\boldsymbol{Lx}$ to vary quite broadly. Hence, we expect that $\boldsymbol{x}^{\text{MAP}}_{\text{ridge}}$ will be affected by significant noise. This is consistent with the bias-variance trade off reported earlier [53].

Choosing a different prior can give substantially different results. For example we can set that each element of $\boldsymbol{x}$ is non-negative and that it is distributed according to a Laplace distribution, i.e., $x_j \sim 1\left(x_j \geq 0\right)$ Laplace $\left(0, \frac{1}{2\lambda}\right)$, whose pdf is $p\left(x_j\right) = \frac{\lambda}{2}\exp\left(-\frac{\lambda}{2}|x_j|\right)$ for $x_j \geq 0$ and 0 otherwise. Such distribution is characterized by a much "fatter" tail than the Gaussian since $\exp(-|x|)$ decays to 0 more slowly than $\exp\left(-\frac{x^2}{2\lambda}\right)$. This choice gives the following negative log-likelihood (multiplied by two) [70,72]

$$\updownarrow_{\text{Lasso}}(\boldsymbol{x}, \lambda) = \sum_{n=1}^{N}\frac{r'^2_n}{\sigma'^2_n} + \sum_{n=1}^{N}\frac{r''^2_n}{\sigma''^2_n} + \lambda\boldsymbol{x}_1 \tag{18}$$

We have used the subscript Lasso to emphasize that this negative log-likelihood is connected to the Lasso regression since the corresponding MAP estimate is obtained from

$$\boldsymbol{x}^{\text{MAP}}_{\text{Lasso}} = \underset{\boldsymbol{x}\geq 0}{\text{argmin}}\updownarrow_{\text{Lasso}}(\boldsymbol{x}, \lambda) \tag{19}$$

We note that many other penalties may be found depending on the particular prior information one wishes to assign to the DRT.

2.3. A Hierarchical Bayesian DRT Analysis

In this section we will relax the hypothesis that the regularization level is uniform across all timescales and concurrently we will seek to keep a minimal amount of tunable parameters. First, the scalar λ will be replaced by the vector $\boldsymbol{\lambda}$. Second, we will assume that $\boldsymbol{\lambda}$ is a random vector (endowed with its own prior). We note that this approach is hierarchical in nature since the prior on $\boldsymbol{x}$ depends on $\boldsymbol{\lambda}$, which in turn has its own prior. The goal will be to determine the pdf of both $\boldsymbol{x}$ and $\boldsymbol{\lambda}$. As above the experimental data $\boldsymbol{Z}_{\exp} = \boldsymbol{Z}'_{\exp} + i\boldsymbol{Z}''_{\exp}$ is given and the starting point of the analysis is Bayes theorem [67,71,73]:

$$p\left(\boldsymbol{x}, \boldsymbol{\lambda}|\, \boldsymbol{Z}'_{\exp}, \boldsymbol{Z}''_{\exp}\right) p\left(\boldsymbol{Z}'_{\exp}, \boldsymbol{Z}''_{\exp}\right) = p(\boldsymbol{x}, \boldsymbol{\lambda}) p\left(\boldsymbol{Z}'_{\exp}, \boldsymbol{Z}''_{\exp}|\, \boldsymbol{x}\right) \tag{20}$$

We stress again that the main conceptual difference between (10) and (20) is that we consider also the pdf of $\boldsymbol{\lambda}$, see Fig. 1. Up to a multiplicative constant, (20) can be rewritten as [67]

$$p\left(\boldsymbol{x}, \boldsymbol{\lambda}|\, \boldsymbol{Z}'_{\exp}, \boldsymbol{Z}''_{\exp}\right) \propto p(\boldsymbol{x}, \boldsymbol{\lambda}) p\left(\boldsymbol{Z}'_{\exp}, \boldsymbol{Z}''_{\exp}|\, \boldsymbol{x}\right) \tag{21}$$

Furthermore, the following holds:

$$p(\boldsymbol{x}, \boldsymbol{\lambda}) = p(\boldsymbol{x}|\, \boldsymbol{\lambda}) p(\boldsymbol{\lambda}) \tag{22}$$

We take a Gaussian prior on $\boldsymbol{x}$, that is $\boldsymbol{L}\boldsymbol{x} \sim \mathcal{N}\left(0, \boldsymbol{\Lambda}^{-1}\right)$ where $\boldsymbol{\Lambda} = \text{diag}(\boldsymbol{\lambda})$ [ii] to obtain [68]

$$p(\boldsymbol{x}|\, \boldsymbol{\lambda}) \propto \mathbb{1}(\boldsymbol{x} \geq 0) \prod_{j=1}^{M'}\sqrt{\frac{\lambda_j}{2\pi}} \exp\left(-\frac{\lambda_j}{2}(\boldsymbol{L}\boldsymbol{x})^2_j\right) \tag{23}$$

We note that, in contrast with (13), in (23) the expected level of noise on the qth derivative depends on the timescale.

We choose that all entries of $\boldsymbol{\lambda}$ are independent and identically distributed with a set of parameters, generically identified by a vector $\boldsymbol{\xi}$, so that

$$p(\boldsymbol{\lambda}) = \prod_{j=1}^{M'}\mathbb{1}\left(\lambda_j \geq 0\right) p_{\text{HP}}\left(\lambda_j, \boldsymbol{\xi}\right) \tag{24}$$

where $p_{\text{HP}}\left(\lambda_j, \boldsymbol{\xi}\right)$ is a given prior pdf. Using an analogous procedure as the one that leads to (16), we can obtain the hyper parametric negative log-likelihood (multiplied by two)

$$\updownarrow^{\text{hyper}}_{\text{ridge}}(\boldsymbol{x}, \boldsymbol{\lambda}) = \sum_{n=1}^{N}\frac{r'^2_n}{\sigma'^2_n} + \sum_{n=1}^{N}\frac{r''^2_n}{\sigma''^2_n} + \boldsymbol{L}_\lambda\boldsymbol{x}^2 + \sum_{j=1}^{M'}\left(-2\ln p_{\text{HP}}\left(\lambda_j, \boldsymbol{\xi}\right) - \ln\lambda_j\right) \tag{25}$$

[ii] If $\boldsymbol{L}$ the derivative matrix of order p, then it has dimensions $M' \times M$ and the matrix $\boldsymbol{\Lambda} = \text{diag}(\boldsymbol{\lambda})$ is diagonal such that $(\boldsymbol{\Lambda})_{ij} = \lambda_i\delta_{ij}$ where δ_{ij} is the Kronecker delta. We note that if a finite different approach is used then $M' = M - q$, and if radial basis functions are used then $M' = M$.

where $\boldsymbol{L}_\lambda = \boldsymbol{\Lambda}^{1/2}\boldsymbol{L}$ and $\boldsymbol{\Lambda} = \mathrm{diag}\left(\lambda_1, \lambda_2, \ldots, \lambda_{M'}\right)$. The MAP vector $\boldsymbol{x}^{\mathrm{MAP}}$ may be determined along with the hyper parameters $\boldsymbol{\lambda}$ as the solution of the following problem [74]:

$$\min_{\boldsymbol{x}\geq 0 \& \boldsymbol{\lambda}\geq 0} \updownarrow^{\mathrm{hyper}}_{\mathrm{ridge}}(\boldsymbol{x}, \boldsymbol{\lambda}) \tag{26}$$

The solution of (26) can be obtained by alternate minimization with respect to $\boldsymbol{\lambda}$ and $\boldsymbol{x}$ [68]. In other words we use the iterative routines outlined in Algorithm 1, where the loop is interrupted when either a given maximum number of iterations is reached or the increment on the iterated $\boldsymbol{x}$ is less than a prescribed amount.

Algorithm 1 Algorithm for the alternate minimization used in hierarchical DRT.

$k = 0$; continue_loop = 1; $\boldsymbol{x}_0 = \boldsymbol{x}_{\mathrm{guess}}$;
while continue_loop
$\boldsymbol{\lambda}^* = \underset{\boldsymbol{\lambda}\geq 0}{\mathrm{argmin}} \updownarrow^{\mathrm{hyper}}_{\mathrm{ridge}}(\boldsymbol{x}_k, \boldsymbol{\lambda})$
$\boldsymbol{x}_{k+1} = \underset{\boldsymbol{x}\geq 0}{\mathrm{argmin}} \updownarrow^{\mathrm{hyper}}_{\mathrm{ridge}}\left(\boldsymbol{x}, \boldsymbol{\lambda}^*\right)$
if ($k > k_{\max}$ & $\boldsymbol{x}_{k+1} - \boldsymbol{x}_k <$ eps) continue_loop = 0; end
$k = k + 1$
end

It is important to note that the Algorithm 1 is particularly convenient if the optimal $\boldsymbol{\lambda}^*$ can be obtained analytically from

$$\nabla_{\boldsymbol{\lambda}} \updownarrow^{\mathrm{hyper}}_{\mathrm{ridge}}(\boldsymbol{x}_k, \boldsymbol{\lambda}) = 0 \tag{27}$$

by choosing a suitable hyperprior [68,75]. As derived in Appendix B exponential, Gaussian, gamma, and inverse gamma, give the analytical expressions reported in Table 1. Therefore, the computational effort necessary to solve one hyperparametric iteration is similar to that of solving a single ridge regression minimization (17), which is a standard quadratic programming problem.

Table 1
List of the hyperprior distributions used and corresponding λ_j^* computed by solving (27). Note that for the exponential and Gaussian priors $\boldsymbol{\xi} = \zeta$, while for the gamma and inverse gamma hyperpriors $\boldsymbol{\xi} = \left(\zeta, \beta\right)^{\mathrm{T}}$.

Hyperprior	Distribution $p_{\mathrm{HP}}\left(\lambda_j, \boldsymbol{\xi}\right) \propto$	Optimal λ_j^*
Exponential	$\mathbb{1}\left(\lambda_j \geq 0\right) \frac{\zeta}{2} \exp\left(-\frac{\zeta}{2}\lambda_j\right)$	$\frac{1}{(\boldsymbol{L}\boldsymbol{x})_j^2 + \zeta}$
Gaussian	$\mathbb{1}\left(\lambda_j \geq 0\right) \sqrt{\frac{\zeta}{2}} \exp\left(-\frac{\zeta}{2}\lambda_j^2\right)$	$\frac{\sqrt{(\boldsymbol{L}\boldsymbol{x})_j^4 + 8\zeta} - (\boldsymbol{L}\boldsymbol{x})_j^2}{4\zeta}$
Gamma	$\mathbb{1}\left(\lambda_j \geq 0\right) \lambda_j^{\frac{\beta}{2}-1} \exp\left(-\frac{\zeta}{2}\lambda_j\right)$	$\frac{\beta - 1}{(\boldsymbol{L}\boldsymbol{x})_j^2 + \zeta}$
Inverse Gamma	$\mathbb{1}\left(\lambda_j \geq 0\right) \lambda_j^{-\frac{\beta}{2}-1} \exp\left(-\frac{\zeta}{2\lambda_j}\right)$	$\frac{\sqrt{(\beta+1)^2 + 4\zeta(\boldsymbol{L}\boldsymbol{x})_j^2} - (\beta+1)}{2(\boldsymbol{L}\boldsymbol{x})_j^2}$

3. Results and Discussion

3.1. Synthetic Experiments

We tested the hyperparametric approach using carefully controlled stochastic experiments. In particular, we chose equivalent circuits with analytical DRT and compared the outcome of the stochastic experiments as a function of the hyperprior employed, the regularization level, and the error structure. We used both

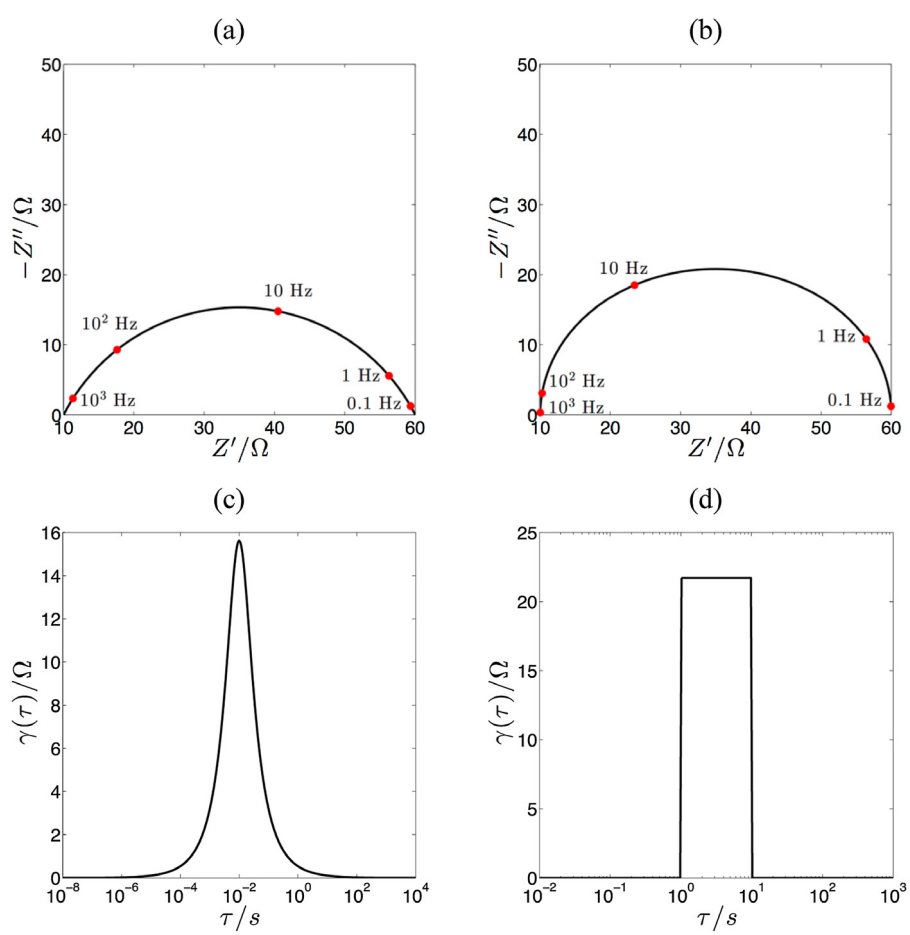

Fig. 2. Noiseless impedance of the ZARC circuit, panel (a), and corresponding DRT, panel (c). Piecewise linear DRT, panel (d), and corresponding noiseless impedance, panel (b).

Table 2
Parameters of the "exact" impedance models.

Parameter	Numerical Value
R_∞	10 Ω
R_{ct}	50 Ω
R_p	50 Ω
τ_0	0.01 s
τ_1	1 s
τ_2	10 s
ϕ	0.7

smooth and piecewise constant $\gamma(\ln\tau)$ to illustrate the proposed method. In particular, we employed a ZARC model consisting of a CPE element placed in parallel to a resistor with resistance R_{ct} and in series with a resistor R_∞. Its impedance is given by [60]

$$Z(f) = R_\infty + \frac{R_{ct}}{1 + (i2\pi f\tau_0)^\phi} \tag{28}$$

with R_∞, R_{ct}, $\tau_0 > 0$ and $0 < \phi \le 1$. The $\gamma(\tau)$ of (28) has the following analytical form [60]

$$\gamma(\tau) = \frac{R_{ct}}{2\pi} \frac{\frac{\tau}{\tau_0}\sin((1-\phi)\pi)}{\cosh\left(\phi \ln\left(\frac{\tau}{\tau_0}\right)\right) - \cos((1-\phi)\pi)} \tag{29}$$

The corresponding $Z(f)$ and $\gamma(\tau)$ are shown in Fig. 2 where the parameters used for plotting are listed in Table 2.

We also simulated a piecewise constant DRT characterized by the following impedance response:

$$Z(f) = R_\infty + \frac{R_p}{\ln\frac{\tau_2}{\tau_1}}\left(\ln\left(1 - \frac{i}{2\pi f\tau_1}\right) - \ln\left(1 - \frac{i}{2\pi f\tau_2}\right)\right) \tag{30}$$

with $\tau_1, \tau_2 > 0$ and $R_p > 0$. The corresponding $\gamma(\tau)$ has the following analytical form

$$\gamma(\tau) = \frac{R_p}{\ln\frac{\tau_2}{\tau_1}}\left(H(\tau - \tau_1) - H(\tau - \tau_2)\right) \tag{31}$$

where $H(\tau)$ is the Heaviside function.

3.1.1. The ZARC Circuit

We first studied the ZARC circuit simulated using the parameters listed in Table 2. We chose 5 points per decade from 10^{-2} to 10^6 Hz. Furthermore, we considered simulated experiments obtained using three different error models, namely

$$Z_{exp}^{abs}(f) = Z(f) + \varepsilon_{abs}\left|Z(f)\right|\left(E' + iE''\right) \tag{32a}$$

$$Z_{exp}^{prop}(f) = Z(f) + \varepsilon_{prop}\left(Z'(f)E' + iZ''(f)E''\right) \tag{32b}$$

$$Z_{exp}^{unif}(f) = Z(f) + \varepsilon_{unif}\left(E' + iE''\right) \tag{32c}$$

where $E', E'' \sim \mathcal{N}(0,1)$. The subscript exp indicates that these are stochastic experiments and the superscript abs, prop, and unif reflect different error structures. These error structures are also connected to distinct weights $\frac{1}{\sigma_n'^2}$ and $\frac{1}{\sigma_n''^2}$ in the likelihoods as shown by (11) and (12). In order to roughly ensure the same error

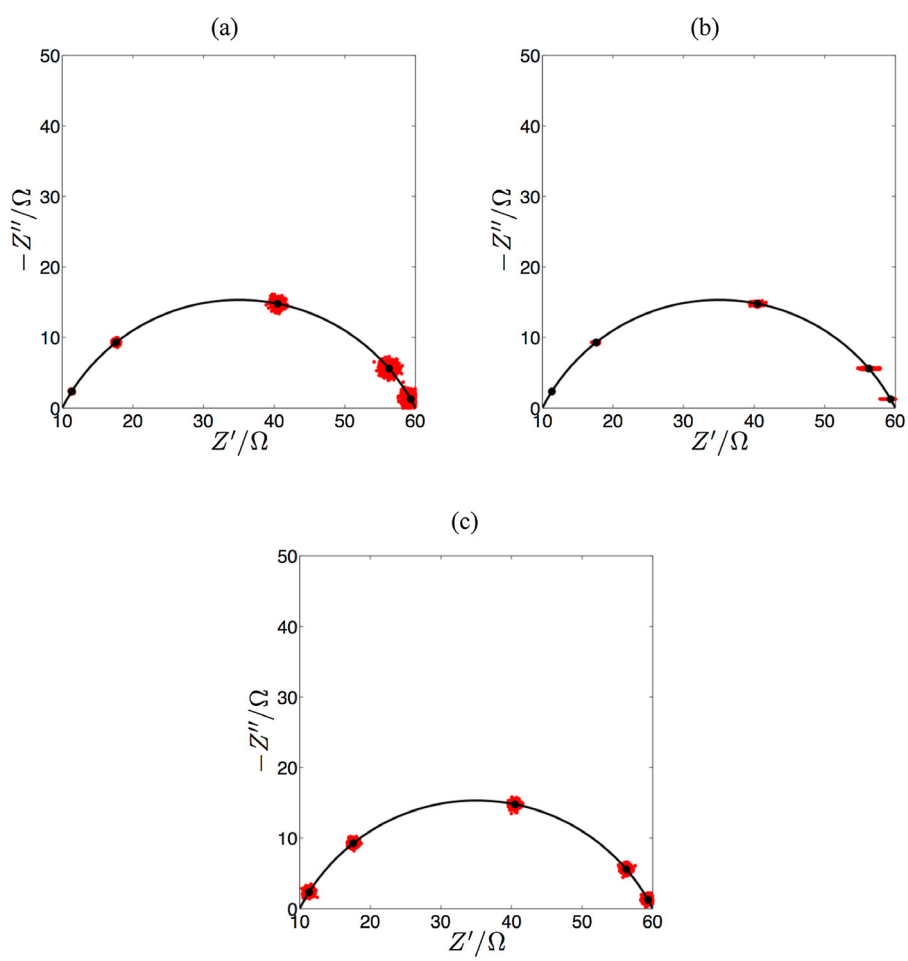

Fig. 3. Outcome of the 1000 stochastic impedance experiments (ZARC circuit) shown at the select frequencies highlighted in Fig. 2, panel (a). The error models (32 a), (32 b), and (32 c), correspond to panels (a), (b) and (c) respectively.

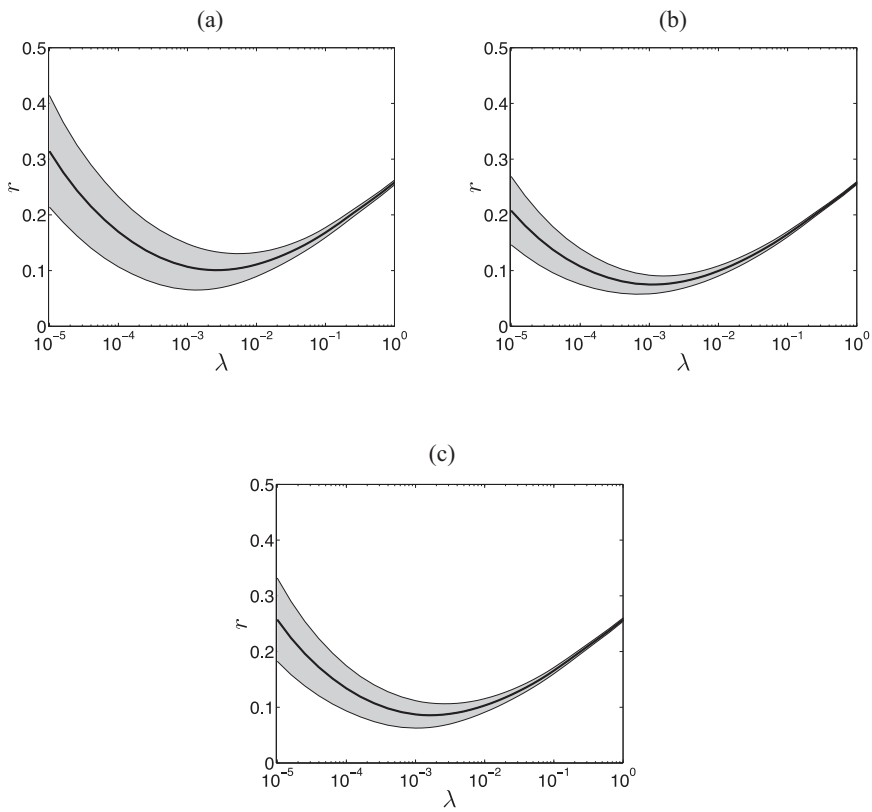

Fig. 4. Average distance between the exact and deconvolved DRT shown with its one σ confidence band. Panel (a), (b), and (c) correspond to abs, Re-Im and uniform errors respectively.

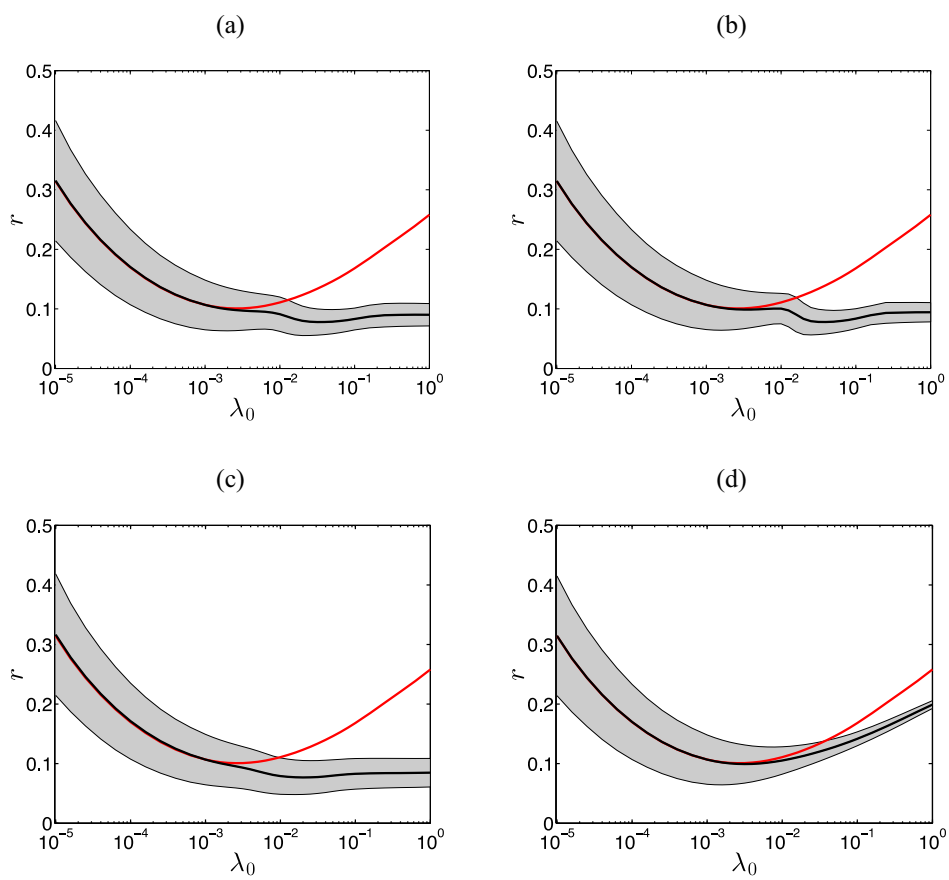

Fig. 5. Average normalized error as a function of λ_0 (Table 3) and corresponding one σ band for error model (32 a). Each panel corresponds to a different hyperprior pdf, namely exponential (a), Gaussian (b), gamma (c), and inverse gamma (d).

level we set that $\varepsilon_{\text{abs}} \left\langle \left|Z\right| \right\rangle = \varepsilon_{\text{prop}} \left(\left\langle \left|Z'\right| \right\rangle + \left\langle \left|Z''\right| \right\rangle \right) = \varepsilon_{\text{unif}}$, where the bracket $\langle\cdot\rangle$ indicates the average over the considered frequency span. The outcome of 1000 stochastic experiments with $\varepsilon_{\text{abs}} = \frac{1}{100}$, is shown as a cloud of points in Fig. 3 where only the data at the frequencies highlighted in Fig. 2 (a) are reported. One immediately notices that different structures give qualitatively distinct simulated EIS experiments, which in turn give different stochastic realizations around the noiseless impedance $Z(f)$. As reported in Fig. 3 (a), the cloud of points for $Z_{\text{exp}}^{\text{abs}}(f)$ is wider at lower frequencies because $\left|Z(f)\right|$ is large. Additionally, the simulated impedance realizations are scattered in a circle around the noiseless $Z(f)$ since the errors on the real and imaginary part have an identical distribution. The $Z_{\text{exp}}^{\text{prop}}(f)$, Fig. 3 (b), gives a cloud of points that is preferentially elongated along the real axis. This is because $Z'(f)$ is greater than $Z''(f)$. Additionally, as the real part of $Z(f)$ decreases with frequency the size of the cloud diminishes. The $Z_{\text{exp}}^{\text{unif}}(f)$ is instead characterized by a constant error level. Hence, as shown in Fig. 3 (c), the cloud of simulated experiments scattered around the exact $Z(f)$ have all identical radius.

In order to test the performance of the various algorithms for a sufficiently large number of simulated experiments, we first drew 1000 realizations of (32). Subsequently, we performed regression by solving (8) for each stochastic experiment and compared each obtained DRT with an exact DRT. We repeated the same procedure for regularization parameters λ in the range from 10^{-5} to 1. In order to uniform the presentation, we chose a piecewise linear basis with $N = M$ [53]. Furthermore, we set $\boldsymbol{\Omega}'$ and $\boldsymbol{\Omega}''$ in (8) to be identity matrices and $\boldsymbol{L}$ to be the second order differentiation matrix. We monitored the residual as a function of λ. The latter is defined as the distance between the exact DRT taken at discrete locations $\boldsymbol{\gamma} = (\gamma(\ln\tau_1), \gamma(\ln\tau_2), \ldots, \gamma(\ln\tau_m))^{\text{T}}$ and the regressed DRT $\boldsymbol{x}^{\text{MAP}}$ normalized with respect to the norm of the exact $\boldsymbol{\gamma}$:

$$r(\lambda) = \frac{\left\|\boldsymbol{x}^{\text{MAP}} - \boldsymbol{\gamma}\right\|}{\left\|\boldsymbol{\gamma}\right\|} \tag{33}$$

We note that studying this residual implies a blended frequentist-Bayesian approach because we use $\boldsymbol{x}^{\text{MAP}}$ as a point estimate rather than a random variable. This is, however, motivated by the need of testing the performance of the method rather than just finding the Bayesian posterior given one single simulated experiment.

Fig. 4 shows the mean r's (solid lines) along with their variability (grey regions) obtained using the three model experiments (32). It is clear that all r's are qualitatively similar. At small λ the errors are large and characterized by large variance (large grey area). For large λ 's the observed errors are equally large but they are characterized by lower variance (small grey area) indicating that the other component, the bias, is large. For a suitable bias-variance tradeoff obtained at intermediate λ the relative error is minimized. The same conclusions can be obtained for the three error models (32) as shown in the panels of Fig. 4. These results are consistent with the conclusions obtained in previous work [53].

We also deconvolved the same synthetic data using the hyperparametric approach. In other words, we solved (26), where in (25) we set $\sigma_n' = \sigma_n'' = 1$, and where we used hyperpriors with exponential, Gaussian, gamma ($\beta = 1.5$), and inverse gamma ($\beta = 1$) pdfs as described in Appendix B and listed in Table 1. These correspond to the λ_j^* 's listed in Table 3. We varied the nominal regularization level

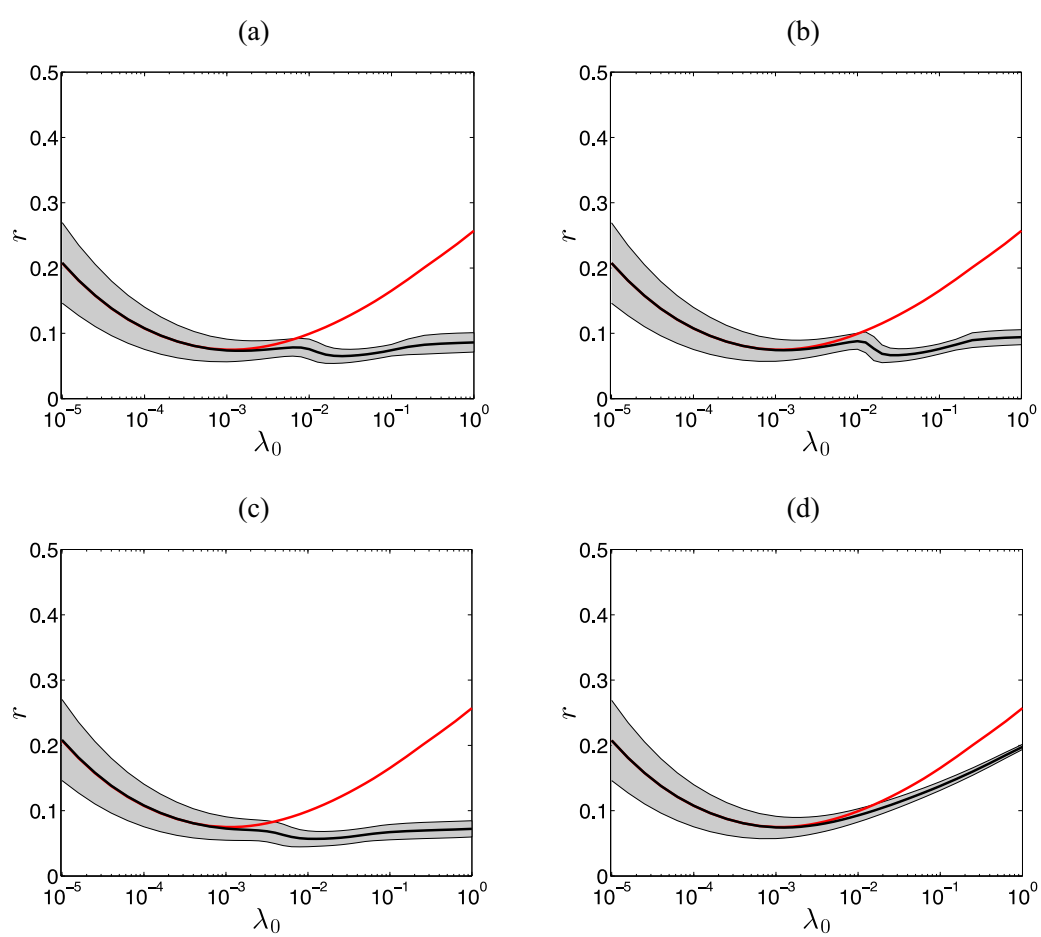

Fig. 6. Plots analogous to Fig. 5 for synthetic experiments sampled with (32 b).

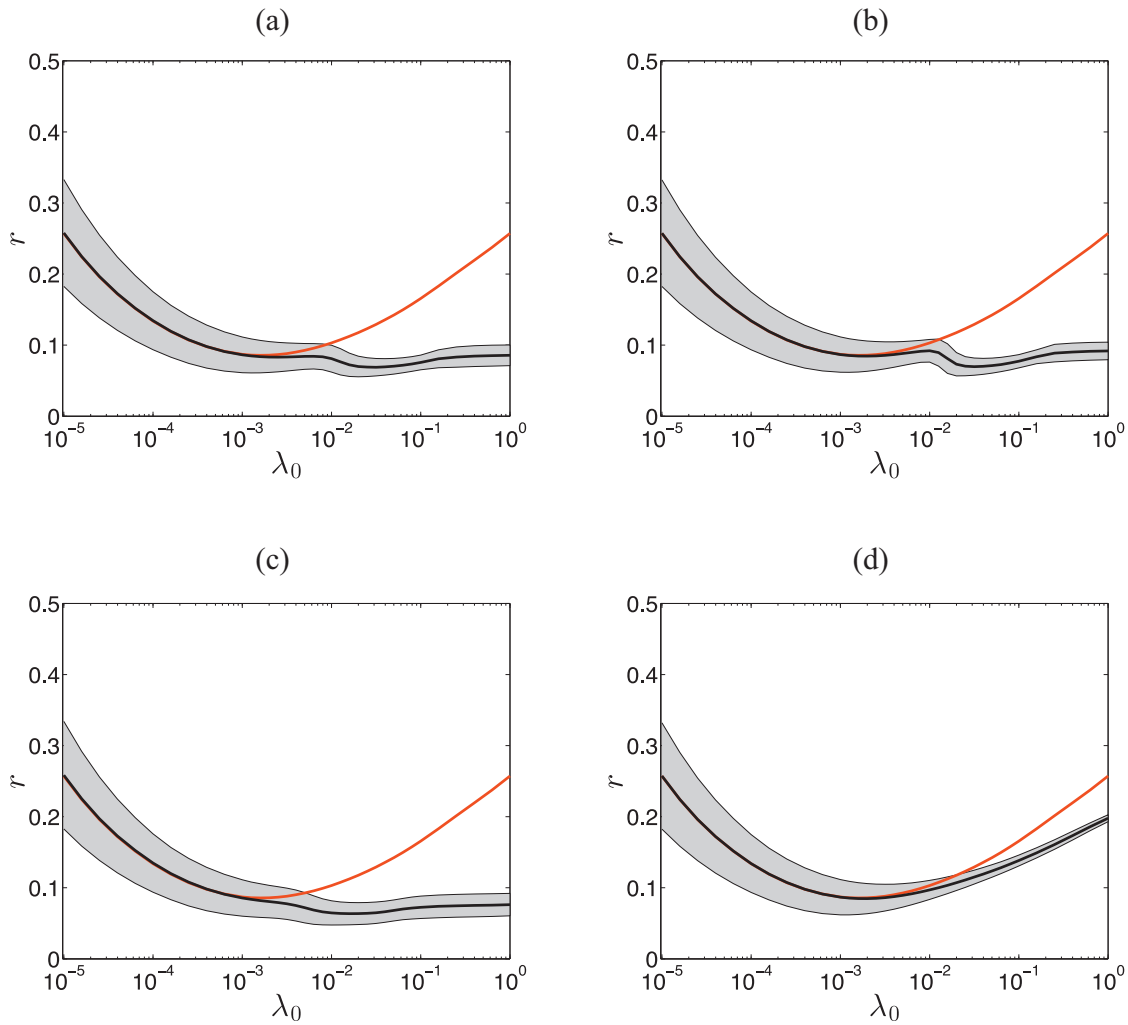

Fig. 7. Plots analogous to Fig. 5 for synthetic experiments obtained using (32 c).

$\lambda_0 = \lim_{(\boldsymbol{Lx})_j^2 \to 0} \lambda_j^*$, in the range from 10^{-5} to 1 .We then plotted $r(\lambda_0)$, the normalized distance between the regressed DRT and the exact $\gamma(\ln\tau)$, in Fig. 5, Fig. 6, and Fig. 7, where each figure displays analogous quantities for the different error structures. Interestingly, we obtained that while exponential, Gaussian, gamma, and inverse gamma hyperpriors behave in a similar manner as the regular ridge DRT for low λ_0 's, they all lead to an abatement of the total error at large λ_0 's. In other words, the hyperparametric approach applied to the ZARC element does not lead to variance reduction (low regularization levels) but it effectively damps out the effect of bias (high regularization levels). Even for large λ_0 's the bias decreases and yet the variance is comparable to the optimal DRT. In fact, the size of the grey confidence band is not as small as in the regular ridge method, where the variance nears zero. In essence, the effect of the hyperparameters is to widen the λ_0 's optimality window. As shown in Fig. 5, Fig. 6, and Fig. 7 this seems to apply to all the error models considered and the hyperparametric approach appears to consistently lower the errors due to bias. In order to complete this discussion, we also illustrate a qualitative comparison of the average MAP estimates against the regular ridge DRT for large regularization parameters λ. As shown in Fig. 8 (a) and (b), for $\lambda = 10^{-1}$ and $\lambda = 1$ respectively, the γ estimated using ridge regression has smaller derivatives and a more pronounced bias with respect to the underlying DRT. Under the same conditions the hyperparametric γ, Fig. 8 (c) and (d), is closer to the original DRT showing better bias control. It is important to note that, while the results shown in Fig. 8 were computed using the simulated experiment (32 c), we expect to draw similar conclusions for all the synthetic experiments (32).

Among the various regularization hyperpriors, the ones with Gaussian, exponential, and gamma pdfs appear to perform better than the inverse gamma. Incidentally, we note that both the exponential and Gaussian λ_j^* 's depend on a single parameter,

Table 3
Asymptotes of λ_j^* for the hyperprior pdfs studied and corresponding conditions in square brackets.

Hyperprior	λ_0	λ_∞
Exponential	$\frac{1}{\zeta}$	$\frac{1}{(\boldsymbol{Lx})_j^2}$
	$\left[(\boldsymbol{Lx})_j^2 \ll \zeta\right]$	$\left[(\boldsymbol{Lx})_j^2 \gg \zeta\right]$
Gaussian	$\frac{1}{\sqrt{2\zeta}}$	$\frac{1}{(\boldsymbol{Lx})_j^2}$
	$\left[(\boldsymbol{Lx})_j^4 \ll 4\zeta\right]$	$\left[(\boldsymbol{Lx})_j^4 \gg 4\zeta\right]$
Gamma	$\frac{\beta-1}{\zeta}$	$\frac{\beta-1}{(\boldsymbol{Lx})_j^2}$
	$\left[\left(\beta-1\right)(\boldsymbol{Lx})_j^2 \ll \zeta\right]$	$\left[\left(\beta-1\right)(\boldsymbol{Lx})_j^2 \gg \zeta\right]$
Inverse Gamma	$\frac{\zeta}{\beta+1}$	$\frac{\sqrt{\zeta}}{\left\lvert(\boldsymbol{Lx})_j\right\rvert}$
	$\left[(\boldsymbol{Lx})_j^2 \ll \frac{(\beta+1)^2}{4\zeta}\right]$	$\left[(\boldsymbol{Lx})_j^2 \gg \frac{(\beta+1)^2}{4\zeta}\right]$

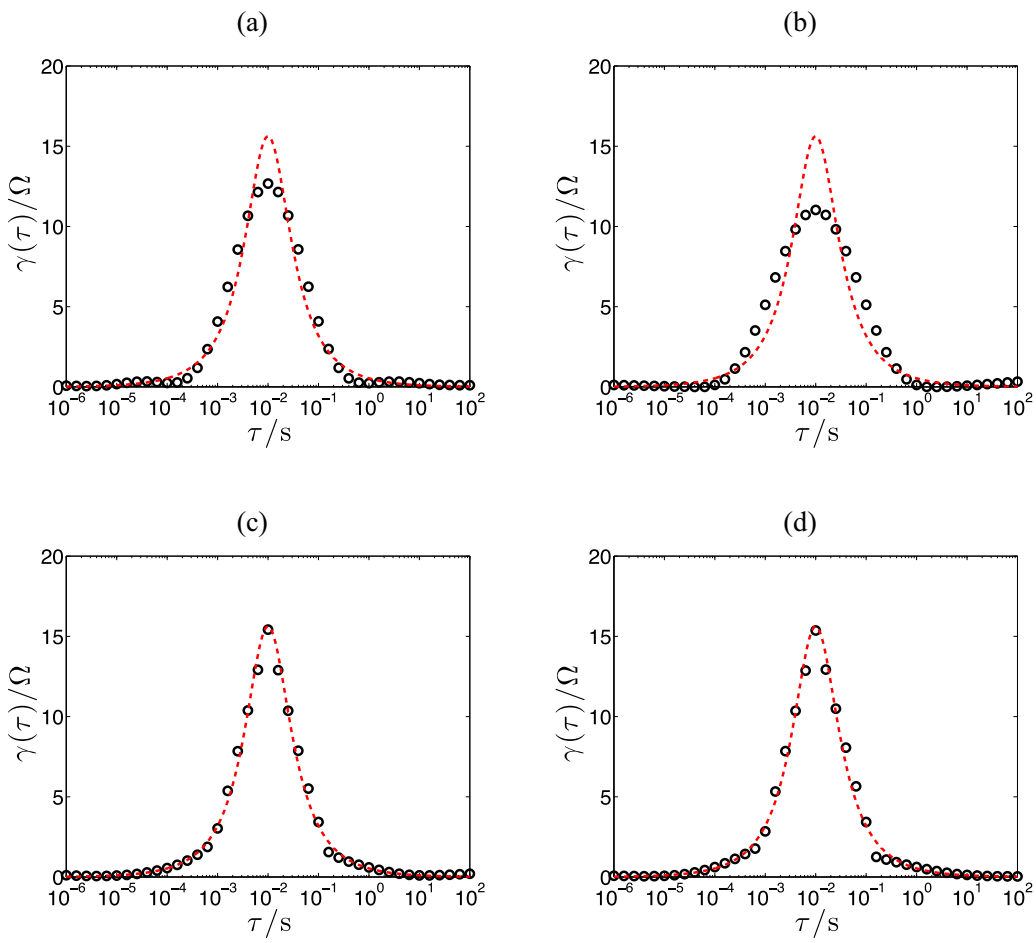

Fig. 8. Panels (a) and (b) show the averaged ridge DRT computed far from the optimum at $\lambda = 10^{-1}$ (a) and $\lambda = 1$ (b). Analogous plots for the hyperparametric case with $\lambda_0 = 10^{-1}$ (c) and $\lambda_0 = 1$ (d).

while the λ_j^* 's obtained using a gamma hyperprior depend on a set of two parameters $\boldsymbol{\xi} = \begin{pmatrix} \zeta & \beta \end{pmatrix}^{\mathrm{T}}$. We also note that, as shown in Appendix B and reported in Table 1, the gamma hyperprior with $\beta = 2$ coincides with its exponential counterpart. Due to its increased tunability and overall good performance, in the remainder of the paper we will focus on the hyperprior with gamma pdf. This choice will allow us to study the relative importance of the derivative and the fixed component of the optimal regularization level. In particular, we can write the optimal λ_j^* (Table 1) as

$$\lambda_j^* = \frac{1}{\frac{1}{\beta-1}(\boldsymbol{L}\boldsymbol{x})_j^2 + \frac{1}{\lambda_0}} \tag{34}$$

where $\lambda_0 = \frac{\zeta}{\beta-1}$ (Table 3). It is clear that the λ_j^* is obtained by weighing the qth derivative term along with the level of nominal regularization λ_0 and that in general $\lambda_j^* \leq \lambda_0$. Further, if qth derivatives are small, i.e., $(\boldsymbol{L}\boldsymbol{x})_j^2 \ll \frac{\beta-1}{\lambda_0}$ (see Table 2), then the expected regularization level is $\lambda_j^* \approx \lambda_0$, giving a regular ridge weight. Conversely, if the derivatives are large, i.e., $(\boldsymbol{L}\boldsymbol{x})_j^2 \gg \frac{\beta-1}{\lambda_0}$, then λ_j^* drops to $\lambda_j^* \approx \frac{\beta-1}{(\boldsymbol{L}\boldsymbol{x})_j^2}$, under-smoothing the solution. This implies that if we fix λ_0, we will be able to switch between nominal and under smoothing by simply modifying β. In particular, $\beta \approx 1$ generally undersmoothes the DRT, giving large variance, while $\beta \gg 1$ leads to results analogous to ridge DRT. For small enough λ_0, the regularization will be such that the term $\frac{1}{\beta-1}(\boldsymbol{L}\boldsymbol{x})_j^2$ is small, again leading to a DRT analogous to the ridge DRT. This is exemplified in Figs. 5–7 (a) & (c) for $\lambda_0 \leq 10^{-2}$.

3.1.2. Circuit with Piecewise Constant DRT

We also performed synthetic experiments with an exact impedance $Z(f)$ given by (30) and shown in the Nyquist form in Fig. 2 (b). This corresponds to the piecewise constant $\gamma(\tau)$ reported in Fig. 2 (d) (we recall that Table 2 contains the parameters used). We stress that we used this type of impedance because its DRT cannot be obtained by ridge regression. In fact, by imposing a penalty on the qth derivative, ridge regression requires that the corresponding DRT is at least differentiable up to order q. We used the model error (32 a) with the frequencies ranging from 10^{-3} to 10^{3}Hz with 20 points per decade. We deconvolved the impedance data using the regular ridge DRT and the hyperparametric method with gamma pdf. Furthermore, we took $\boldsymbol{L}$ to be the 2nd order differentiation matrix. We studied the impact of small, medium, and large relative weights attributed to the derivatives, see (34), by selecting $\beta = 1.01$, 2 and 30 respectively and $\lambda_0 = 0.1$.

As predicted, the regular ridge regression method does not capture faithfully the exact DRT. This is shown in Fig. 9 panel (a) and reproduced for reference in panels (b) and (c). The regular method tends to smooth out the entire DRT by penalizing with equal strength the derivatives at every timescale. Conversely, the hyperparametric approach leads to generally better results. As illustrated in Fig. 9 panels (a) and (b), corresponding to $\beta = 1.01$ and 2 respectively, the exact piecewise constant DRT is recovered quite faithfully. If instead $\beta = 30$, Fig. 9 panel (c), the discontinuity is

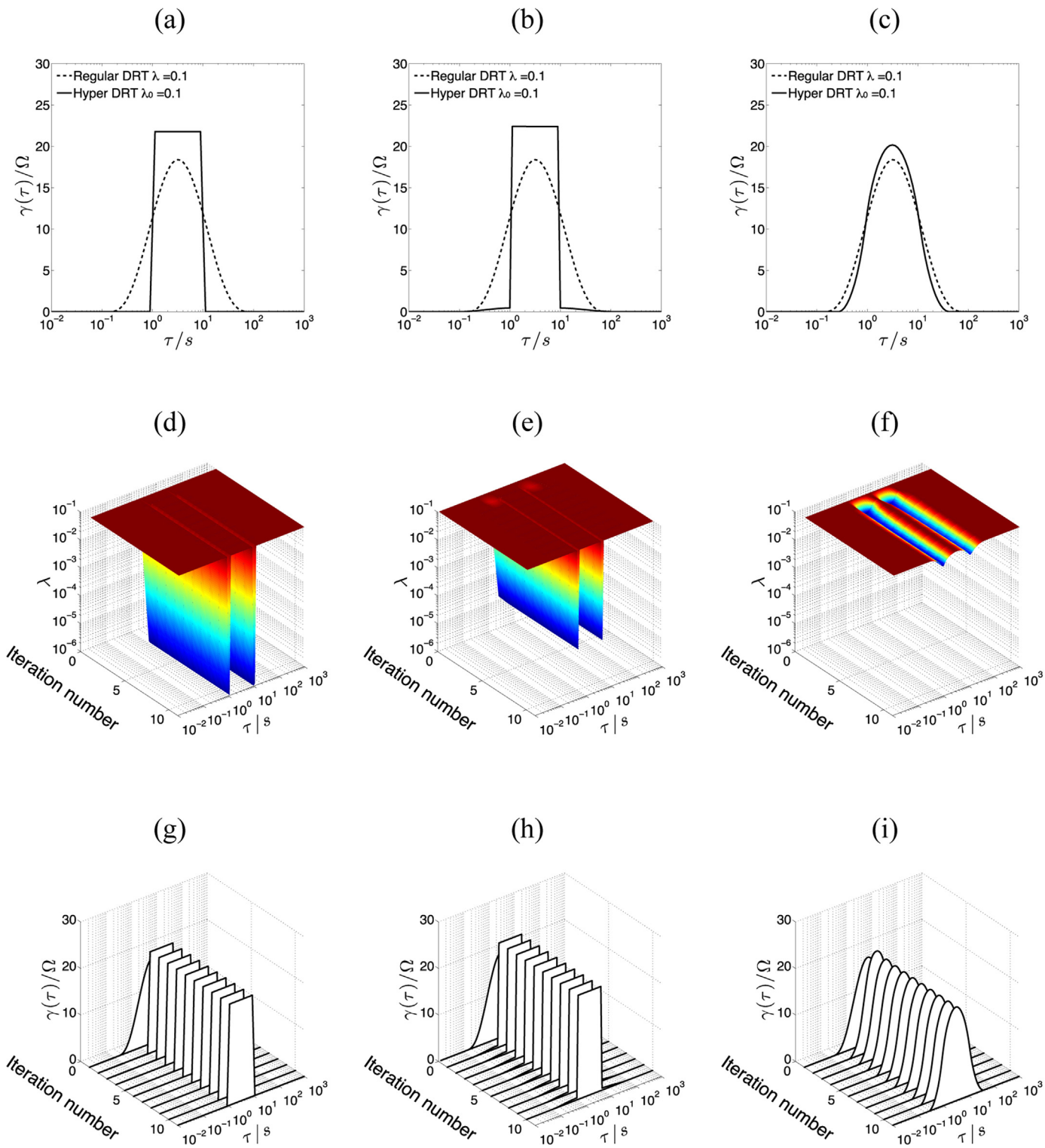

Fig. 9. Piecewise constant DRT recovered by regular and hyperparametric DRT methods (a), (b), and (c). The evolution of the local regularization level $\boldsymbol{\lambda}$ as a function of iteration number (d), (e), and (f) (see Algorithm 1) and the corresponding evolution of the DRT (g), (h), and (i). The first column corresponds to $\beta = 1.01$, the second to $\beta = 2$ and the third to $\beta = 30$.

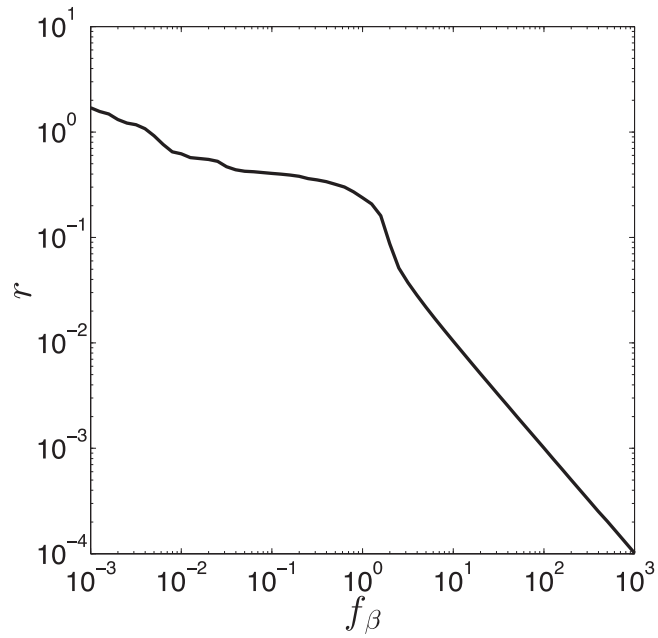

Fig. 10. Plot of r, the normalized distance between hyperparametric DRT and the optimal Re-Im cross validated DRT, as a function of the parameter f_β.

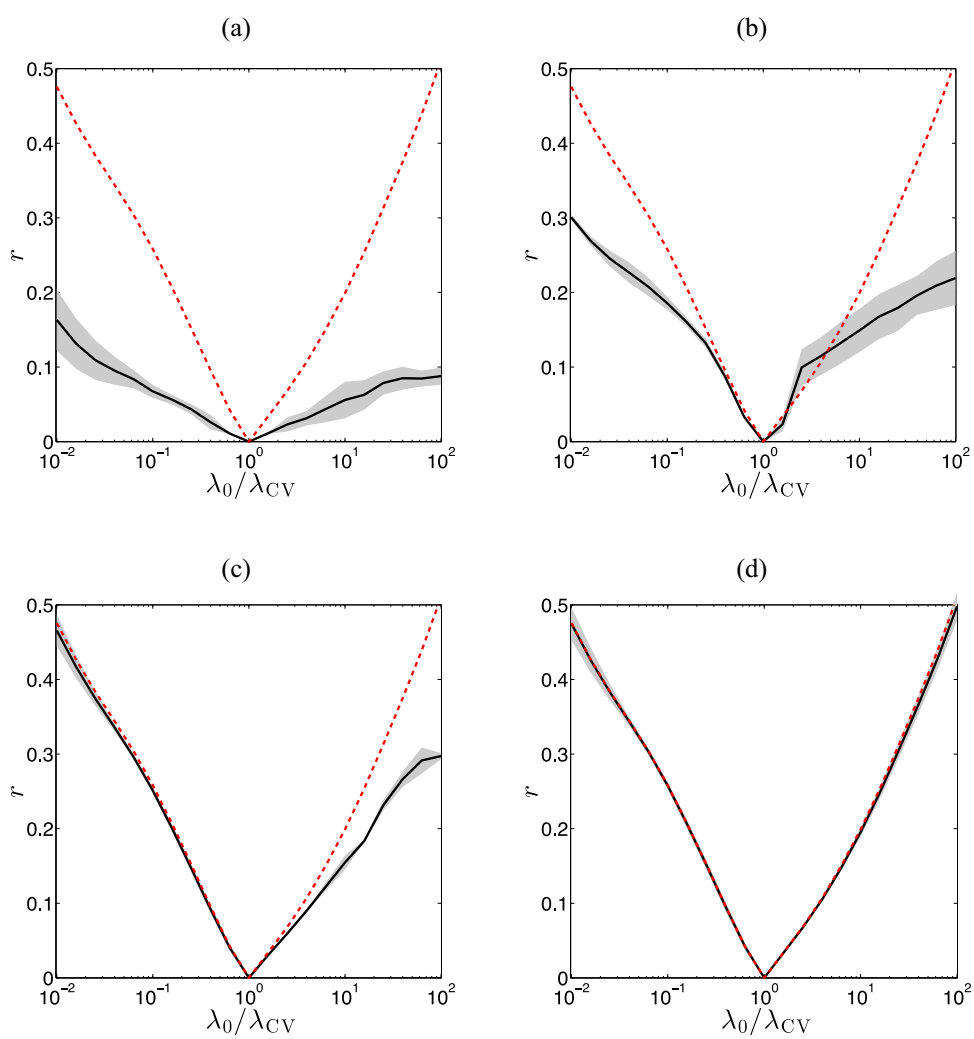

Fig. 11. Normalized distance between the hyperparametric DRT computed at λ_{CV} and the hyperparametric DRT computed at various nominal regularization levels λ_0 with fixed f_β (or β). The case $f_\beta \to \infty$ (ridge DRT) is shown in all plots as a red dashed line. Panels (a), (b), (c), and (d) correspond to $f_\beta = 0.1$, 1, 10, 100 respectively.

not recovered. This behavior can be easily rationalized by looking into (34). If $\beta \approx 1$, the denominator is large since the derivatives' contribution overwhelms the nominal regularization level λ_0, i.e., $\frac{1}{\beta-1}(\boldsymbol{Lx})_j^2 \gg \frac{1}{\lambda_0}$. Since the piecewise constant DRT has large numerical derivatives at the discontinuity, the local regularization level, λ_j^*, is close to zero making it possible to capture the corresponding jump. If β is instead in the intermediate range (here we set $\beta = 2$), then the orders of magnitude of the terms $\frac{1}{\beta-1}(\boldsymbol{Lx})_j^2$ and $\frac{1}{\lambda_0}$ are closer. In turn, this makes the hyperparametric approach less sensitive to the contribution of the derivatives. As illustrated in Fig. 9 panel (b), while the general characteristics of the piecewise constant DRT are recovered, additional features emerge near the discontinuity. If β is even larger, the impact of the derivatives will be further weakened and the hyperparametric method will lead to results close to those obtained by applying regular DRT as shown in Fig. 9 panel (c).

The hyperparametric approach relies on the refinement of the local regularization level, thereby modifying the regressed $\gamma(\ln\tau)$ at each iteration. The evolution of the local regularization level is shown in Fig. 9 panels (d), (e), and (f) for $\beta = 1.01$, 2 and 30 respectively. As explained above, the regularization level declines sharply near the discontinuity for $\beta = 1.01$. This decrease is weaker for $\beta = 2$, and almost absent for $\beta = 30$. If $\beta = 30$, the λ_j^* 's are roughly uniform across the entire time spectrum. Concomitantly, with the regularization level's progression, the $\gamma(\tau)$ will evolve as a function of iteration number as shown in Fig. 9 panels (g), (h), and (i). Starting from the DRT obtained from regular ridge regression, the hyperparametric approach will converge to the final $\gamma(\tau)$ within a few iterations by emphasizing the relevant features.

3.2. Analysis of Lithium – ion battery Experiments

We finally applied the DRT regularization to the Lithium-ion battery (LIB) spectra reported in our earlier publication [53]. We stress that is relevant because, as it is well known, the EIS responses of LIBs are so complicated that they require equivalent circuits with 10 or more parameters. In turn, this has adverse effects on the identifiability of the respective models [76,77]. We note that in contrast with our previous work, we selected the basis elements in (6) to be radial basis functions [78].

As a first step, we screened the regular ridge DRT for all the 12 available spectra and determined for each spectrum the regularization parameter $\bar{\lambda}_{CV}$ by minimizing the Re-Im cross validation. We found that $\lambda = \bar{\lambda}_{CV} + \Delta\lambda_{CV} = 2.2 \cdot 10^{-5} \pm 3.16 \times 10^{-6}$. We then employed the hyperparametric method with gamma hyperprior. We did this in order to study the relative impact of the parameters β and λ_0 in (34) and Table 1. We first rewrote (34) in a more convenient form by defining a new quantity f_β, which normalizes β as follows

$$f_\beta = \frac{1}{\lambda_0} \frac{\beta - 1}{\max\limits_{j=1,\ldots,M'} (\boldsymbol{Lx})_j^2} \tag{35}$$

This in turn allows us to rewrite (34) as

$$\lambda_j^* = \frac{\lambda_0}{\frac{1}{f_\beta} \frac{(\boldsymbol{Lx})_j^2}{\max\limits_{j=1,\ldots,M'} (\boldsymbol{Lx})_j^2} + 1} \tag{36}$$

where it is clear that the factor f_β directly tunes the relative importance of the derivative term. In particular, if f_β is small, then the local derivative will decrease the regularization level. Conversely, for large f_β the derivatives have little contribution ($\lambda_j^* \approx \lambda_0$) and ultimately we recover ridge regression.

As a first step we analyzed the impact of β at fixed $\lambda = \bar{\lambda}_{CV}$. We then performed regularization as a function f_β and calculated the normalized distance $r\left(f_\beta\right)$ between the DRT computed using the regular ridge method and the hyperparametric approach. This is defined as $r\left(f_\beta\right) = \frac{\left\|\gamma_{hyper}\left(f_\beta, \lambda_{opt}, \ln\tau\right) - \gamma_{CV}(\ln\tau)\right\|}{\|\gamma_{CV}(\ln\tau)\|}$, where $\gamma_{CV}(\ln\tau)$ is the ridge DRT obtained for the cross-validated penalty. Its average is shown in Fig. 10. For small f_β ($\beta \approx 1$) the derivatives have a large impact on the λ_j^* 's. In turn, the DRT obtained from the hyperparametric method deviates from the ridge DRT as shown by a rather large r value. By increasing the f_β, this difference decreases and the hyperparametric approach leads to DRTs closer to those obtained from simple regularization. In fact, $r\left(f_\beta\right)$ tends to 0 as f_β goes to infinity.

As shown in Section 3.1.1, the hyperparametric method with a carefully chosen β works well for λ_0 's beyond the optimum $\bar{\lambda}_{CV}$. Thus, as a second step, we explored the effect of λ_0 given β. We stress that, while an analytical solution of the DRT is not available, we took each individual hyperparametric DRT calculated at λ_{CV} as the reference condition. We then fixed β and computed the normalized distance between the DRT computed at λ and and the DRT at $\bar{\lambda}_{CV}$ with the same β, namely $r(\lambda) = \frac{\left\|\gamma_{hyper}\left(f_\beta, \lambda, \ln\tau\right) - \gamma_{hyper}\left(f_\beta, \lambda_{CV}, \ln\tau\right)\right\|}{\|\gamma_{hyper}\left(f_\beta, \lambda_{CV}, \ln\tau\right)\|}$. We repeated this computation for $f_\beta = 0.1$, 1, 10 and 100 as shown in Fig. 11 panels (a), (b), (c), and (d) respectively. In Fig. 11 we average $r(\lambda)$ (solid line) along with the square root of the variance computed across the 12 experiments. Consistently with the intuition gained from expression (36) and as shown in Fig. 11 (d), for large f_β (or β) the hyperparametric approach leads to solutions very similar to those obtained using the regularized DRT method (dashed line). Conversely, for smaller f_β (β progressively closer to 1) the $r(\lambda)$ is generally characterized by a smaller slope, see Fig. 11 panels (a), (b), and (c). In particular, the slope decreases as a function of f_β. This indicates that the closer β is to 1, the less sensitive is the method to variations of λ_0.

In order to further support these insights, we computed the DRTs obtained using the regular and hyperparametric approach at $\bar{\lambda}_{CV}$ and at λ_0 much greater than the optimum. Panels (a) and (b) of Fig. 12 show the DRTs computed at $\lambda_0 = \bar{\lambda}_{CV}$ and $\lambda_{off-CV} = 100 \cdot \bar{\lambda}_{CV}$ respectively. As displayed in Fig. 12 (a), if λ_0 increases, the typical

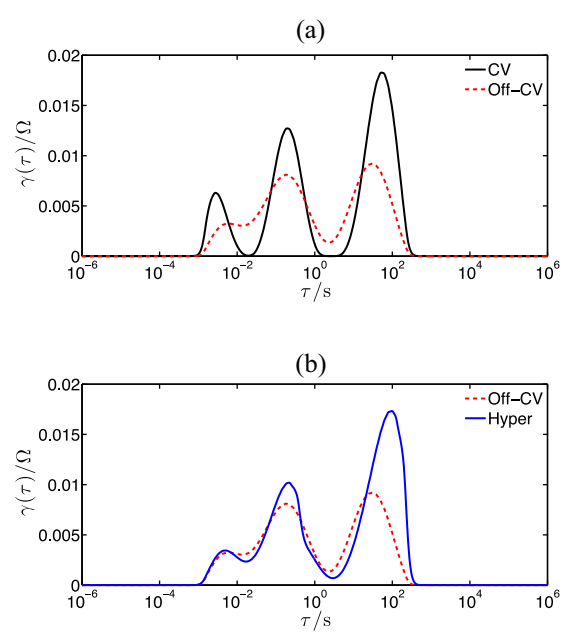

Fig. 12. Panel (a) shows the ridge DRT obtained at the Re-Im cross validated regularization level λ_{CV} (CV) and at $100 \cdot \lambda_{CV}$ (off-CV). Panel (b) shows ridge (off-CV) and hyperparametric DRT (hyper), which are both computed at $\lambda_0 = 100 \cdot \lambda_{CV}$.

ridge DRT is significantly flattened, such that the peaks are hardly visible. In contrast, the hyperparametric DRT with $\lambda_0 = 100 \cdot \bar{\lambda}_{CV}$ and $f_\beta \approx 10$ is capable of detecting the relevant DRT peaks with only minimal bias with respect to the ridge DRT at $\lambda_0 = \bar{\lambda}_{CV}$ as shown in Fig. 12 (b).

4. Conclusions

In this article we have shown that the ridge regularization as applied to DRT can be obtained using Bayesian statistics. We used this insight to expand the classical ridge/Tikhonov methodology and develop hierarchical DRT methods. In doing that, we developed ways to adjust the levels of regularization in a timescale dependent fashion.

We employed both synthetic and real experiments to evaluate the performance of the hierarchical DRT in comparison to traditional ridge/Tikhonov DRT and we illustrated that the hyperparametric DRT is able to mitigate the effect of the bias found at large nominal regularization levels. In this regard, the hyperparametric approach is more robust in comparison to traditional ridge regression, allowing a reduced sensitivity to bias at high regularization levels. We were able to show this for real battery data, where even at high regularization levels the estimated DRTs are comparable to those obtained using optimized regularization levels by cross-validation. Furthermore we also showed, by means of a set of stochastic experiments, that the hyperparametric DRT is capable of faithfully capturing discontinuities in the DRT spectrum.

Lastly, we stress that understanding the DRT analysis in a Bayesian context opens up further opportunities for future development of this technique. By using informative priors and perhaps even different formulations, the Bayesian approach may provide alternative ways of examining EIS data via the DRT approach.

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Appendix A. Appendix A

Here we explain how (4) and (5) can be obtained from the hypothesis that the underlying physical processes are relaxations. We start by noting that (1) can be written as

$$v(t) = z(t) * i(t) \tag{A.1}$$

where the * sign indicates the convolution operation defined as $z(t) * i(t) = \int_{-\infty}^{\infty} z(t')\, i(t-t')\, dt'$. Physically this means that a current input $i(t)$ will result in a voltage $v(t)$, which is cross-correlated with $i(t)$ via the transfer function $z(t)$. The $z(t)$ encapsulates an array of physical and chemical phenomena occurring in the system. It is plausible that $z(t)$ is a relaxation, such that for a current impulse (e.g. a input blip current) the voltage will eventually be damped out, relaxing monotonically to zero. The $z(t)$ can be rewritten as the sum of decaying exponentials (relaxations) with timescales $\tau_1 < \tau_2 < \ldots < \tau_K$

$$z(t) = a_0 \delta(t) + \sum_{k=1}^{K} \frac{a_k}{\tau_k} H(t) e^{-t/\tau_k} \tag{A.2}$$

where $H(t)$ is the Heaviside step function, defined as $H(t) = 1$ if $t \geq 0$ and $H(t) = 0$ if $t < 0$, and where $a_k \geq 0$. We note that the term first term on the right hand side $a_0 \delta(t)$ encapsulates the $t = 0$ response (an instantaneous resistance). In other words, a portion of the response will be purely resistive and it will not decay but will turn instantenously on and off dependending on the presence of the input. We note that this is the response corresponds to a timescale $\tau_0 \to 0$ because

$$\lim_{\tau_0 \to 0} \frac{a_0}{\tau_0} H(t) e^{-t/\tau_0} = a_0 \delta(t) \tag{A.3}$$

By taking Fourier transform of (A. 2) one may obtain the impedance of the system, i.e.:

$$Z_{DRT}(f) = \mathcal{F}[z(t)](f) = a_0 + \sum_{k=1}^{K} \frac{a_k}{1 + i2\pi f \tau_k} \tag{A.4}$$

Formally, if we (a) set that $a_k = g_k \Delta\tau$ is an infinitesimally small quantity, (b) let $K \to \infty$, (c) allow the τ_k to span the entire real line, and (d) set $a_0 = R_\infty$, then we obtain (4) and (5) from

$$Z_{DRT}(f) = R_\infty + \sum_{k=1}^{K} \frac{g_k \Delta\tau}{1 + i2\pi f \tau_k} \approx R_\infty + \int_0^\infty \frac{g(\tau)}{1 + i2\pi f \tau} d\tau \tag{A.5}$$

Even more formally, we could have expressed $z(t)$ as the following integral of decaying exponentials:

$$z(t) = R_\infty \delta(t) + \int_0^\infty \frac{g(\tau)}{\tau} H(t) \exp\left(-\frac{t}{\tau}\right) d\tau \tag{A.6}$$

This would have equivalently lead to (4) and (5). We note that relation (A. 6) may actually be further extended. For example, if, in analogy with damped RLC circuits, we had assumed that the response is composed by a relaxation and a harmonic component, we would have obtained the following expression:

$$\begin{aligned} z(t) = R_\infty \delta(t) &+ \int_0^\infty \int_0^\infty \gamma_{\sin}(\tau, \tau') H(t) \exp\left(-\frac{t}{\tau}\right) \\ &\sin\left(2\pi \frac{t}{\tau'}\right) d\tau d\tau' + \int_0^\infty \int_0^\infty \gamma_{\cos}(\tau, \tau') H(t) \\ &\exp\left(-\frac{t}{\tau}\right) \cos\left(2\pi \frac{t}{\tau'}\right) d\tau d\tau' \end{aligned} \tag{A.7}$$

It is clear from the discussion that we may postulate many more DRT-type response functions. Lastly we must note that using expression (A. 6) or equivalently (4) and (5) has numerous advantages, for example we can write that [79]

$$\gamma(\ln\tau) = -\frac{1}{\pi}\left[Z''\left(\frac{1}{2\pi} e^{-\ln\tau + \frac{i\pi}{2}}\right) + Z''\left(\frac{1}{2\pi} e^{-\ln\tau - \frac{i\pi}{2}}\right)\right] \tag{A.8}$$

In other words, given $\gamma(\ln\tau)$ we can compute Z using (4); conversely, given Z we can obtain $\gamma(\ln\tau)$ thanks to (A. 8).

Appendix B. Appendix B

We start from (25) and consider 4 hyper-priors characterized by exponential, Gaussian, gamma, and inverse gamma pdf's.

B.1. Exponential Hyperprior

If we take that the elements of $\boldsymbol{\lambda}$ are independent and identically distributed (i.i.d.) and that all have the same exponential pdf, then

$$p_{HP}(\boldsymbol{\lambda}, \boldsymbol{\xi}) = \left(\frac{\zeta}{2}\right)^{M'} \prod_{j=1}^{M'} \mathbb{1}(\lambda_j \geq 0) \exp\left(-\frac{\zeta}{2}\lambda_j\right) \tag{B.1}$$

with $\boldsymbol{\xi} = \zeta$, this leads to

$$\updownarrow_{\text{ridge}}^{\text{hyper}}(\boldsymbol{x}, \boldsymbol{\lambda}) = \sum_{n=1}^{N} \frac{r'^2_n}{\sigma'^2_n} + \sum_{n=1}^{N} \frac{r''^2_n}{\sigma''^2_n} + \left\|\boldsymbol{L}_\lambda \boldsymbol{x}\right\|^2 + \sum_{j=1}^{M'} \left(\zeta\lambda_j - \ln\lambda_j\right) \tag{B.2}$$

which can be minimized with respect to $\boldsymbol{\lambda}$ for $\nabla_{\boldsymbol{\lambda}} \updownarrow_{\text{ridge}}^{\text{hyper}} (\boldsymbol{x}_k, \boldsymbol{\lambda}) = 0$, i.e.,

$$\lambda_j = \frac{1}{(\boldsymbol{L}\boldsymbol{x})_j^2 + \zeta} \tag{B.3}$$

B.2. Gaussian Hyperprior

If the λ_j are i.i.d. and their underlying pdf are Gaussians with variance $\frac{1}{\zeta}$, i.e.,

$$p_{\text{HP}}\left(\boldsymbol{\lambda}, \boldsymbol{\xi}\right) = \left(\frac{\zeta}{2}\right)^{\frac{M'}{2}} \prod_{j=1}^{M'} \mathbb{1}\left(\lambda_j \geq 0\right) \exp\left(-\frac{\zeta}{2}\lambda_j^2\right) \tag{B.4}$$

where $\boldsymbol{\xi} = \zeta$, then we obtain

$$\updownarrow_{\text{ridge}}^{\text{hyper}} (\boldsymbol{x}, \boldsymbol{\lambda}) = \sum_{n=1}^{N} \frac{r'^2_n}{\sigma'^2_n} + \sum_{n=1}^{N} \frac{r''^2_n}{\sigma''^2_n} + \left\|\boldsymbol{L}_\lambda \boldsymbol{x}\right\|^2 + \sum_{j=1}^{M'} \left(\zeta\lambda_j^2 - \ln\lambda_j\right) \tag{B.5}$$

the minimum may be found by computing the zeros of

$$\frac{\partial \updownarrow_{\text{ridge}}^{\text{hyper}} (\boldsymbol{x}_k, \boldsymbol{\lambda})}{\partial \lambda_j} = (\boldsymbol{L}\boldsymbol{x})_j^2 + 2\zeta\lambda_j - \frac{1}{\lambda_j} = 0 \tag{B.6}$$

where the only positive root is given by

$$\lambda_j^* = \frac{\sqrt{(\boldsymbol{L}\boldsymbol{x})_j^4 + 8\zeta} - (\boldsymbol{L}\boldsymbol{x})_j^2}{4\zeta} \tag{B.7}$$

B.3. Gamma Hyperprior

For a gamma pdf with $\boldsymbol{\xi} = \left(\zeta, \beta\right)^T$ $(\beta \geq 1)$, i.e.,

$$p_{\text{HP}}(\boldsymbol{\lambda}) = \prod_{j=1}^{M'} \mathbb{1}\left(\lambda_j \geq 0\right) \lambda_j^{\frac{\beta}{2}-1} \exp\left(-\zeta\frac{\lambda_j}{2}\right) \tag{B.8}$$

we get instead that

$$\updownarrow_{\text{ridge}}^{\text{hyper-exp}} (\boldsymbol{x}, \boldsymbol{\lambda}) = \sum_{n=1}^{N} \frac{r'^2_n}{\sigma'^2_n} + \sum_{n=1}^{N} \frac{r''^2_n}{\sigma''^2_n} + \left\|\boldsymbol{L}_\lambda \boldsymbol{x}\right\|^2 + \sum_{j=1}^{M'} \left(\zeta\lambda_j - \left(\beta - 1\right)\ln\lambda_j\right) \tag{B.9}$$

And from

$$\frac{\partial \updownarrow_{\text{ridge}}^{\text{hyper}} (\boldsymbol{x}_k, \boldsymbol{\lambda})}{\partial \lambda_j} = (\boldsymbol{L}\boldsymbol{x})_j^2 + \zeta - \frac{\beta - 1}{\lambda_j} = 0 \tag{B.10}$$

we obtain that

$$\lambda_j^* = \frac{\beta - 1}{(\boldsymbol{L}\boldsymbol{x})_j^2 + \zeta} \tag{B.11}$$

B.4. Inverse Gamma Hyperprior

Finally, if $p_{\text{HP}}(\boldsymbol{\lambda})$ is an inverse gamma pdf with $\boldsymbol{\xi} = \left(\zeta, \beta\right)^T$ $(\beta > 0)$

$$p_{\text{HP}}(\boldsymbol{\lambda}) = \prod_{j=1}^{M'} \mathbb{1}\left(\lambda_j \geq 0\right) \lambda_j^{-\frac{\beta}{2}-1} \exp\left(-\frac{\zeta}{2\lambda_j}\right) \tag{B.12}$$

then we obtain

$$\updownarrow_{\text{ridge}}^{\text{hyper}} (\boldsymbol{x}, \boldsymbol{\lambda}) = \sum_{n=1}^{N} \frac{r'^2_n}{\sigma'^2_n} + \sum_{n=1}^{N} \frac{r''^2_n}{\sigma''^2_n} + \left\|\boldsymbol{L}_\lambda \boldsymbol{x}\right\|^2 + \sum_{j=1}^{M'} \left(\frac{\zeta}{\lambda_j} + \left(\beta + 1\right)\ln\lambda_j\right) \tag{B.13}$$

Taking

$$\frac{\partial \updownarrow_{\text{ridge}}^{\text{hyper}} (\boldsymbol{x}_k, \boldsymbol{\lambda})}{\partial \lambda_j} = (\boldsymbol{L}\boldsymbol{x})_j^2 - \frac{\zeta}{\lambda_j^2} + \frac{\beta + 1}{\lambda_j} = 0 \tag{B.14}$$

gives the following positive solution:

$$\lambda_j^* = \frac{\sqrt{\left(\beta + 1\right)^2 + 4\zeta(\boldsymbol{L}\boldsymbol{x})_j^2} - \left(\beta + 1\right)}{2(\boldsymbol{L}\boldsymbol{x})_j^2} \tag{B.15}$$

Lastly, we note that all computed likelihoods are in agreement with (25) up to an additive constant.

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